Enumerative geometry of K3 surfaces

Based on a lecture series by Qaasim Shafi.

December 2023

Enumerative geometry starts off from the very classical problem of counting lines in a variety. We follow two parallel stories — first for a K3 surface, and then for projective space with additional boundary conditions. The count of rational curves on a K3 surface is given by the Yau–Zaslow formula in terms of a modular form, and the counts of higher genus curves is determined by the counts of rational curves. Abstracting the intersection-theoretic arguments shows that in fact we often do not count curves, but instead get 'generalised curve counts' called *Gromov–Witten invariants* which are better behaved under deformations. This deformation invariance is exploited when there are imposed tangency conditions on the count, to degenerate the variety and reduce the problem to a combinatorial count. Throughout, we see how intuition from the K3 count feeds into this general theory, in particular we are able to 'guess' the relationship between counts of rational curves and those of higher genus curves.

§0.1 About. This course on enumerative geometry was a part of a Winter School for early career researchers, organised by the UK Algebraic Geometry Network. The lectures were delivered in-person in the University of Warwick, and were transcribed by Parth Shimpi. Parts of this course are based on a lecture by Rahul Pandharipande (`https://www.youtube.com/watch?v=TBoonBCDRa8`). These notes have undergone several amendments and are not a verbatim recall of the lectures, therefore discretion is advised when using them. They are available online at `https://pas201.user.srcf.net/documents/2023-ukag-enum-geom.pdf`.

All errors and corrections should be communicated to by email to `pas201@srcf.net`.

§1 Rational curves on K3 surfaces

How many lines on a smooth hypersurface $X \subset \mathbb{P}^3$ of degree d? If $d = 2$, then the surface is a quadric (isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$) and is covered by lines — in particular there are two sets transverse rulings. If $d = 3$, then the surface is cubic and famously contains 27 lines — we have lost almost all the lines! More generally, all surfaces of degree $d \leqslant 3$ are rational and hence contain plenty of rational curves. On the other hand if $d \geqslant 5$, it can be shown that X has no lines and no rational curves of any degree.

What about $d = 4$? This case plays a transitional role similar to the one played by elliptic curves for the arithmetic problem of counting $\mathbb{Q}$-points on curves. Suppose X is defined by a degree 4 polynomial f_4. By definition, a degree e rational curve in X is a map $\mathbb{P}^1 \to X$ defined by three homogeneous polynomials $p_0, \dots, p_3$ of degree e satisfying $f_4(p_0, \dots, p_3) = 0$. Hence we may try to naïvely count the dimension of the space of degree e rational curves in X as

$$\overbrace{(\underbrace{4e+4}_{\text{[Each } p_i \text{ has } e+1 \text{ coefficients]}} - \underbrace{1}_{\text{[Projectivise]}} - \underbrace{3}_{\text{[dim Aut}\mathbb{P}^1\text{]}})}^{\dim\{\text{degree } e \text{ rational curves in } \mathbb{P}^1\}} - \overbrace{(\underbrace{4e+1}_{\text{[A degree } 4e \text{ homogeneous polynomial should identically vanish]}})}^{\dim\{\text{constraints to lie on } X\}} = -1.$$

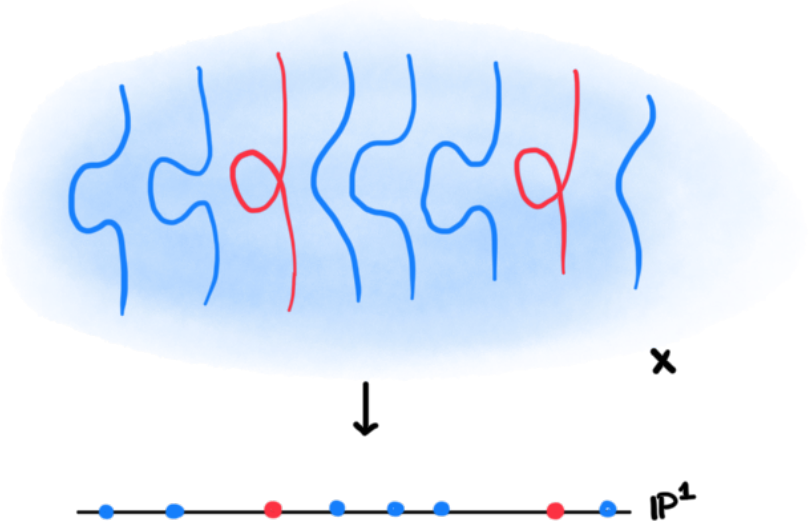
Thus the dimension count suggests there are no rational curves. However, one can very explicitly find rational curves by showing that X intersects a tri-tangent plane in a degree 4 curve with three nodes, i.e. a rational curve! Thus some of the constraints in our dimension count do not intersect transversely.

§1.1 What makes degree 4 interesting?. It is because quartic surfaces sit in a family of algebraic varieties that obey the *Calabi–Yau condition.* In these situations, curve counting is special with the possibility of exact solutions in all genera– in dimension one, Calabi–Yau varieties are precisely the elliptic curves and the counts relate to modular forms. Calabi–Yau 3-folds are an active area of current research, with exact counts for genus 0 given by mirror symmetry.

Definition 1.1. An *algebraic K3 surface* X is a smooth projective $\mathbb{C}$-surface that satisfies $H^1(X, \mathcal{O}_X) = 0$ and has trivial canonical class.

The prime example, of course, are quartic hypersurfaces in $\mathbb{P}^3$. All K3 surfaces are diffeomorphic as smooth $\mathbb{R}$-manifolds, with topological Euler characteristic 24.

§1.2 Rational curves via fibrations. It can be shown that algebraic K3 surfaces arise in families indexed by $h = 1, 2, 3, \ldots$ such that any surface in the family h admits a complete linear system of dimension h, giving a map $X \to \mathbb{P}^h$. A general element in this linear system can be shown to have genus h. Hence if $h = 1$ (for example if X is the Kummer surface), this map is an elliptic fibration.



Deleting the singular fibers, we obtain a locally trivial fibration by Ehresmann's lemma and hence we have

$$\sum_{C \text{ a singular fiber}} \chi_{\text{top}}(C) = \chi_{\text{top}}(X) = 24.$$

But singular fibers in this case are rational, hence X has at least 24 rational curves (for simplicity, we assume all curves are reduced, irreducible, and at worst nodal.)

Likewise, if $h = 2$ then the map $X \to \mathbb{P}^2$ is a double cover branched over a sextic curve $C \subset \mathbb{P}^2$. Such a curve has 324 bitangent lines, each of which pulls back to a fiber with two nodes (again rational, since a generic fiber has genus 2). If $h = 3$, $X \hookrightarrow \mathbb{P}^3$ is a quartic hypersurface and the rational elements in the linear system come from tri-tangent planes like we discussed before. There are precisely 3200 such planes.

§1.3 The Yau–Zaslow formula. Write $N_{0,h}$ for the count of genus 0 curves in the linear system associated to a general algebraic K3 surface. The data of h fixes a cohomology class $\beta \in \text{Pic}(X) = H^2(X, \mathbb{Z}) \cap H^{1,1}(X)$ with $\beta^2 = 2h - 2$, and we are counting (stable) curves $\mathbb{P}^1 \to X$ with image in this cohomology class. From the above discussion, the first few values of $N_{0,h}$ are $24, 324, 3200, \ldots$. Yau and Zaslow spotted that these coincide with the coefficients of a well-known modular form, and conjectured the following.

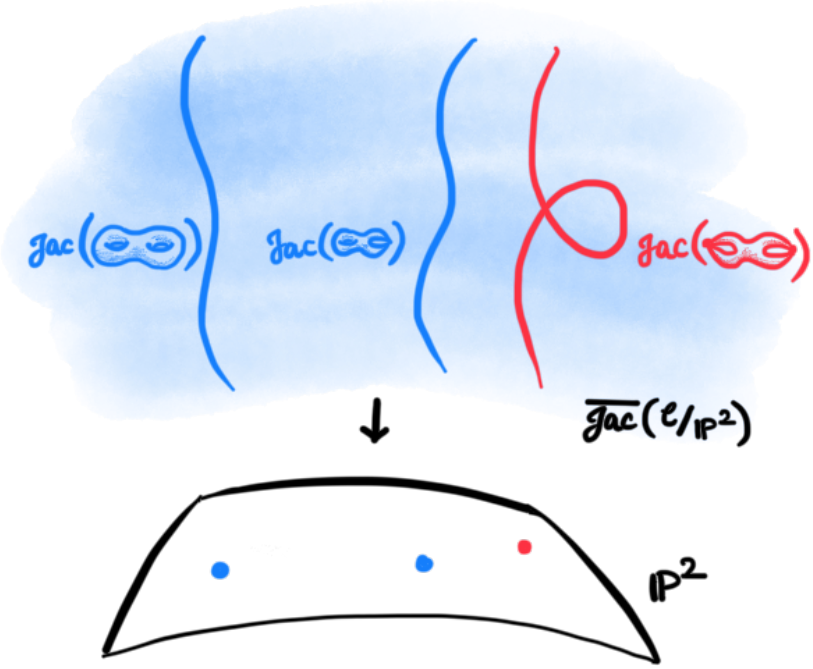
Theorem 1.2 (Yau–Zaslow formula). *Writing* $\Delta(z) = (\prod_{n \geqslant 1}(1 - z^n)^{24})^{-1}$ *for the discriminant modular form, we have the identity*

$$\sum_{h \geqslant 0} N_{0,h} \cdot z^h = \frac{z}{\Delta(z)} = 1 + 24z + 324z^2 + \cdots$$

This was proven independently Beauville and Bryan–Leung. How might one prove this? Inspired by the construction for $h = 1$, we will build another fibration. The base is the linear system $\mathbb{P}^h$, and the total space is the compactified Jacobian of the universal curve $\mathcal{C} \hookrightarrow \mathbb{P}^h$ i.e. the fiber over a smooth curve $C \in \mathbb{P}^h$ is the Jacobian $\mathcal{J}ac(C)$. The space is compactified to include nodal fibers, via a generalised Jacobian construction (for $h = 1$ this is the same fibration we had, since an elliptic curve is its own Jacobian.)

Now the Jacobian $\mathcal{J}ac(C)$ has trivial Euler characteristic if C is smooth or has genuine genus (eg. the first two fibers in the picture), while $\chi_{top}(\mathcal{J}ac\, C) = 1$ if C is rational. Thus we have

$$N_{0,h} = \sum_{\text{rational curves } C} \chi_{top}(\mathcal{J}ac\, C) = \chi_{top}(\overline{\mathcal{J}ac}(\mathscr{C}/\mathbb{P}^h)).$$



Moreover, by a result of Huybrechts we have that $\overline{\mathcal{J}ac}(\mathscr{C}/\mathbb{P}^h)$ is birational to the Hilbert scheme $\mathrm{Hilb}^h(X)$, and the Euler characteristic of the latter has been computed by Göttsche as the coefficient of z^h in $\frac{z}{\Delta(z)}$. A result of Batyrev–Kontsevich shows that two birational Calabi–Yau manifolds have the same Betti numbers, which concludes our argument.

§ 2 Introduction to Gromov–Witten theory

The way we computed the counts on a K3 surface exploited the specific geometry in the situation. We will now describe a more general framework for counting curves, which involves two steps. First, we build a moduli space of curves in X of a fixed type (genus, cohomology class) — if this has dimension zero then we obtain an honest count, otherwise we impose natural conditions by doing intersection theory on this moduli space.

Example 2.1. There are infinitely many conics (i.e. rational curves of degree 2) in $\mathbb{P}^2$. These are parametrised by the moduli space $\mathcal{M}_{0,2}\mathbb{P}^2 = \mathbb{P}^5$, since a conic is determined by a degree 2 homogeneous polynomial in three variables. Thus to obtain a count, we may impose the condition that our curve passes through five fixed points $\{p_1, \ldots, p_5\}$. Each point gives a linear constraint, and hence if the five points are in general position then there is a unique conic passing through them (corresponding to the intersection of the five general hyperplanes.)

As a general principle, the count should not depend on the specific choice of points. To make this rigorous, we replace each hyperplane $H_i = \{\text{conics through } p_i\}$ with its (Chow) homology class $[H_i]$, and then define the intersection to be given by cup products of Poincaré duals

$$[H_1] \cdot \ldots \cdot [H_5] = \int_{[\mathbb{P}^5]} \mathrm{PD}[H_1] \cup \ldots \cup \mathrm{PD}[H_5] = 1.$$

The count then depends only on the cohomology classes of the constraints. Theories of modern enumerative geometry follow this principle.

Remark 2.2. In general, there are two to think about curves in X — via parametrisations (i.e. as maps $C \to X$), or via equations. Thus for curves in $\mathbb{P}^2$ one might think of parametrisations $[z:w] \mapsto [z^2 : zw : w^2]$ or equations $\mathbb{V}(xz - y^2)$. The first approach leads to Gromov–Witten theory, while the latter gives rise to Donaldson–Thomas theory.

Thus to do Gromov–Witten theory, we first fix a genus $g \geqslant 0$ and a cohomology class $\beta \in H^2(X, \mathbb{Z})$, and 'guess' the moduli space

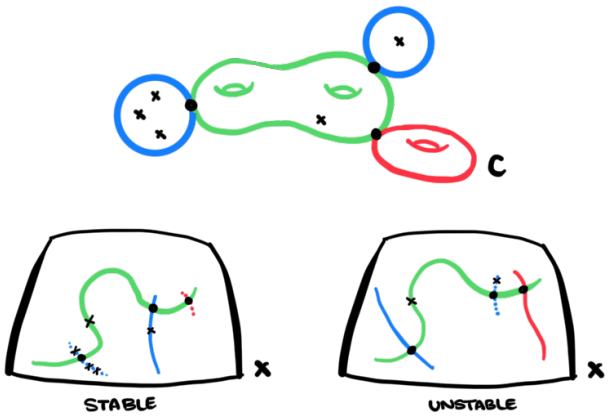
$$\mathcal{M}_g(X, \beta) = \{C \xrightarrow{f} X \mid C \text{ a smooth curve of genus } g, \quad f_*[C] = \beta\}.$$

However, this naïve moduli space is often non-compact (and hence unsuitable for intersection theory.) To get a compactification, we have to allow maps f from nodal curves which may contract components. The well-behaved maps of this kind are called stable (defined later). Our space of interest, now compact, is

$$\overline{\mathcal{M}}_g(X, \beta) = \{C \xrightarrow{f} X \mid C \text{ an at worst nodal curve of genus } g, \quad f \text{ stable}, \quad f_*[C] = \beta\}.$$

To impose constraints, it is also often convenient to consider curves with $n \geqslant 0$ *marked points* $p_1, \ldots, p_n \in C$, and form the moduli space $\overline{\mathcal{M}}_{g,n}(X, \beta)$ of stable maps $f : C \to X$ with prescribed behaviour (e.g. tangency conditions) at $f(p_i)$.

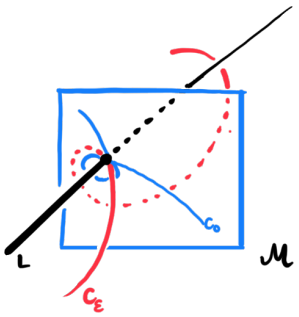
Definition 2.3. Let C be a nodal curve of genus g with n marked points. A map $f : C \to X$ is *stable* if every rational component contracted by f has at least three nodes or marked points, and every contracted elliptic component has at least one node or marked point.



We have succeeded, by obtaining a compact moduli space $\overline{\mathcal{M}}_{g,n}(X, \beta)$. At what cost? We have added way more junk — the complement $\overline{\mathcal{M}}_{g,n}(X, \beta) \setminus \mathcal{M}_{g,n}(X, \beta)$ is often massive and has components of dimension (way) higher than the original space. To make matters worse, $\overline{\mathcal{M}}_{g,n}(X, \beta)$ often does not have a fundamental class — we have nothing to integrate over for intersection theory!

The second of these problems is remedied by introducing a *virtual fundamental class* $[\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\text{vir}}$, if we are lucky this coincides with the fundamental class (this happens, for instance, for $\overline{\mathcal{M}}_{0,n}(\mathbb{P}^N, d)$. If not, we pretend the compactified moduli space of interest $\mathcal{M}$ is cut out by r equations in a smooth ambient space Y of dimension N. The excess junk of dimension $> N - r$ arises precisely because the intersection is not transverse in places, so we perturb these equations to make it transverse and think of $[\mathcal{M}]^{\text{vir}} \in H_{N-r}(\mathcal{M})$ as the fundamental class of this perturbed space.

Example 2.4. Suppose $\mathcal{M} = \mathbb{V}(xz, yz) \subset \mathbb{P}^3$ is the union of a line $L = \mathbb{V}(x, y)$ (the space of interest) and a plane $\mathbb{V}(z)$ (the junk). Perturbing the equations, we obtain $\mathbb{V}(xz - \epsilon y^2, yz - \epsilon xw)$ which is the union of the line L and a twisted cubic $C_\epsilon : [u, v] \mapsto (v^3, \epsilon uv^2, \epsilon^2 u^2 v, \epsilon^3 u^3)$. In the limit, we have a plane cubic $C_0 \subset \mathbb{V}(z)$ and the virtual fundamental class of $\mathcal{M}$ is $[\mathcal{M}]^{\text{vir}} = [L] + [C_0]$.



Then the *Gromov–Witten invariants* are defined as integrals of the form $\int_{[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\mathrm{vir}}}[\text{constraints}]$ where the constraints usually naturally arise from the setting of the problem and live in the same dimension as the virtual fundamental class. Because the moduli space has so much junk and the fundamental class is virtual, these are not really curve counts any more. However, they are better behaved in other ways — they are deformation invariant, and naturally arise in mirror symmetry. And often, they do help compute actual curve counts.

§ 2.1 Gromov–Witten theory of a K3 surface. Let X be a K3 surface and $\beta \in \operatorname{Pic} X$ a primitive class with $\beta^2 = 2h-2$. Rather annoyingly, the virtual fundamental class $[\overline{\mathcal{M}}_g(X,\beta)]^{\mathrm{vir}}$ is zero for all g. This is related to the "-1" dimension count we obtained earlier, though we do not go into the details. The fix here is to define a 'reduced virtual fundamental class' $[\overline{\mathcal{M}}_g(X,\beta)]^{\mathrm{red}}$, which is non-zero and lives in $H_g(\overline{\mathcal{M}}_g(X,\beta))$. In particular for rational curves this class has degree zero, and the curve counts are simply

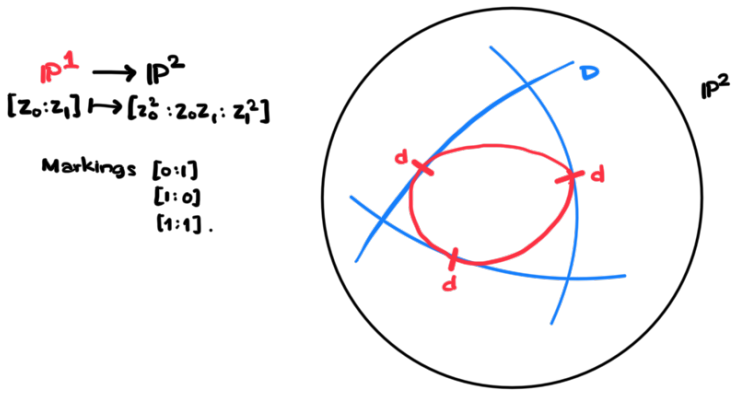
$$N_{0,h} = \int_{[\overline{\mathcal{M}}_g(X,\beta)]^{\mathrm{red}}} 1.$$

For $g > 0$ the curve count is infinite, since the virtual dimension of the moduli space is positive. To get a finite number, we impose a g-dimensional condition λ_g to obtain a count $N_{g,h}$ (this condition is vacuous for $g = 0$.) Roughly speaking, this condition arises from a rank g vector bundle $\mathbb{E}_g \to \overline{\mathcal{M}}_g(X,\beta)$ whose fiber over a stable curve C is given by $H^0(C,\omega_C)$. Thus these constraints are sufficiently natural, and we get meaningful invariants of X.

The Yau–Zaslow formula has a generalisation in higher genera, called the KKV conjecture. This asserts that generating series for $N_{g,h}$ lie in the ring of quasi-modular forms, and was proven by Maulik, Pandharipande, and Thomas.

§ 3 Logarithmic Gromov–Witten theory

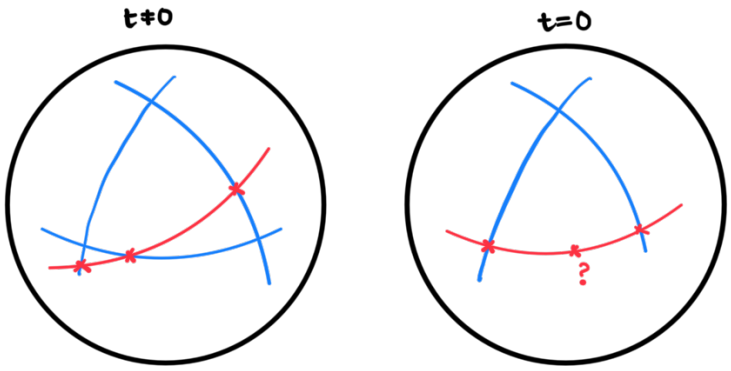
To see how marked points play a role in Gromov–Witten theory, we will now try to answer questions of the flavour 'how many (stable) curves of fixed type in a variety X have prescribed tangency conditions to a given divisor $D \subset X$ with simple normal crossings?' Concretely, let us count degree d plane rational curves that are maximally tangent to each axis. Thus we are looking at degree d maps $f : \mathbb{P}^1 \to \mathbb{P}^2$ such that $f(\mathbb{P}^1) \cap \mathbb{V}(x) = d \cdot f[0:1]$ and likewise for the other markings.



In general, we fix g, n, β as before and α a collection of integers giving the desired tangency of the ith marking to D. Consider the moduli space

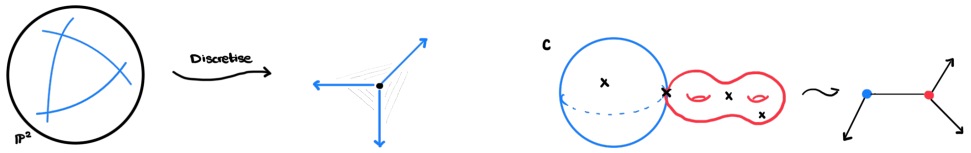
$$\mathcal{M}_{g,\alpha}(X|D,\beta) = \left\{\text{smooth curves } C \xrightarrow{f} X \text{ of genus } g \text{ and degree } \beta \text{ with tangency } \alpha \text{ with } D\right\}.$$

Is this compact? No — as before, we have only allowed smooth curves so we fix that by asking for stable nodal curves instead. The resulting space is still non-compact, because it does not include curves that degenerate into D. For example, consider the curve given by $[z_0 : z_1 : t(z_0 - z_1)]$ as $t \to 0$. In the limit, the curve degenerates into D as shown below.



Thus we need a new fix for this problem. For this, we will need to think torically/tropically, by 'discretising' each of the players X, C, f in the problem.

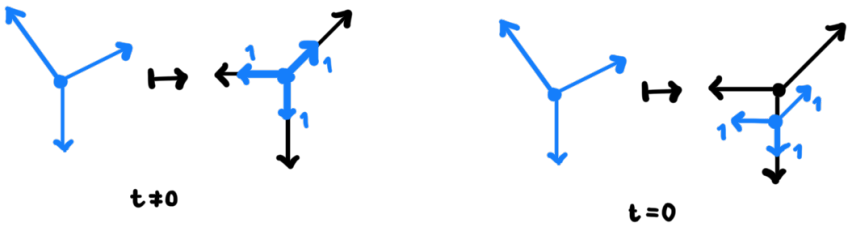
§ 3.1 The tropicalisation game. In general, we follow the following rules: the pair X, D is replaced by the associated toric fan, and the curve C becomes a graph with vertices representing irreducible components, edges representing nodes, and 'floating' edges corresponding to other markings.



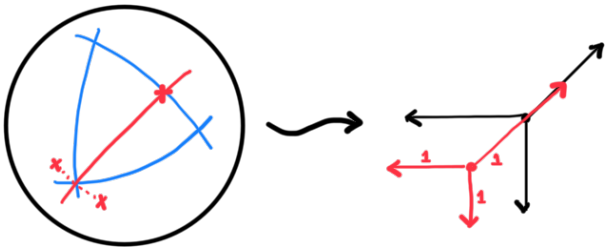
The map $f : C \to X$ is then represented by a linear embedding of the graph, where each vertex maps into the cone of the toric stratum in which the curve generically sits. The floating edges are parallel to the ray which gives the component of D on which the corresponding marked point lies.



Thus for the family of curves $[z_0 : z_1 : t(z_0 - z_1)]$, we have

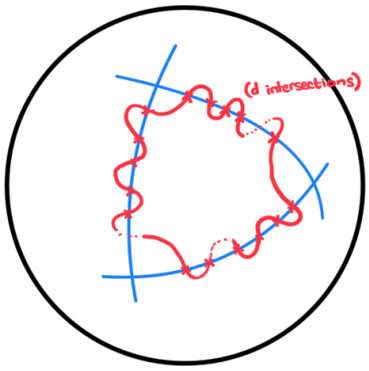


§ 3.2 What's in the moduli space?. Say vertex in the degenerated curve is *balanced* if the sum of all outgoing vectors is zero. To first approximation, we add into our moduli space those stable maps whose discretisation is balanced. For example, the following map is detected.



The resulting space $\overline{\mathcal{M}}^{\log}_{g,\alpha}(X|D,\beta) \supset \mathcal{M}_{g,\alpha}(X|D,\beta)$ is compact, and is called the *moduli space of stable logarithmic maps.* This, as before, does not have a fundamental class but does have a virtual one. Doing intersection theory on the space then gives *logarithmic Gromov–Witten invariants.*

§3.3 Curve counts using log Gromov–Witten theory. To count general curves (and hence bring log Gromov–Witten theory closer to the ordinary one), we can enforce generic tangency conditions with the divisor. For example, a general degree d curve in $\mathbb{P}^2$ looks like



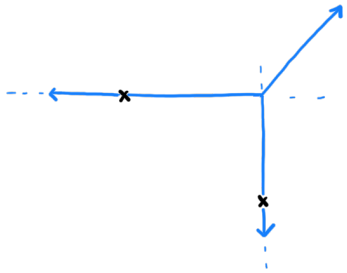
so we impose a generic tangency constraint by considering $\mathbb{P}^1$ with $3d$ marked points. This gives a $3d-1$ dimensional space of generic degree d curves, so by imposing additional conditions (e.g. requiring the curve to pass through $3d-1$ fixed points $p_1,\dots,p_{3d-1}\in\mathbb{P}^2$) we can define the Gromov–Witten invariant

$$N_{0,d}^{\mathbb{P}^2}=\int_{[\overline{\mathcal{M}}_{0,\alpha}(\mathbb{P}^2|D,d)]^{\mathrm{vir}}}[\text{condition to pass through } p_1,\dots,p_{3d-1}].$$

It turns out the problem is sufficiently nice and this is equal to the number of rational degree d curves in $\mathbb{P}^2$ through $3d-1$ generic points. Moreover, it can be computed combinatorially by using the the following theorem and other similar results.

Theorem 3.1 (Mikhalkin). *The number $N_{0,d}^{\mathbb{P}^2}$ is equal to the number of "discretised curves" in the toric fan of $\mathbb{P}^2$ with d unbounded arrows in each divisorial direction, and passing through through $3d-1$ points in $\mathbb{R}^2$.*

Thus for example if $d=1$, there is a unique balanced curve passing through two points fixed points in $\mathbb{R}^2$ with balanced vertices and one unbounded arrow in each direction. This, of course, corresponds to the fact that there is a unique rational curve through two fixed points in $\mathbb{P}^2$.



§4 Counts in higher genus

We have seen that generalised counts of rational curves (i.e. Gromov–Witten invariants for genus 0) on $\mathbb{P}^2$ are related to tropical curve counts. We will now see how this generalises to statements about higher genus curves. In general forming a logarithmic moduli space of higher genus curves is hard, so we first rephrase the genus 0 problem in purely tropical terms.

Recall the goal was to measure $N_{0,d}^{\mathbb{P}^2}=\#\{\text{plane rational curves of degree } d \text{ through } 3d-1 \text{ points}\}$, and we did this by defining a combinatorial gadget. We make this precise.

For this, fix the data $\Delta_d=\{(-1,0),\overset{d}{\dots},(-1,0),(0,-1),\overset{d}{\dots},(0,-1),(1,1),\overset{d}{\dots},(1,1)\}$.

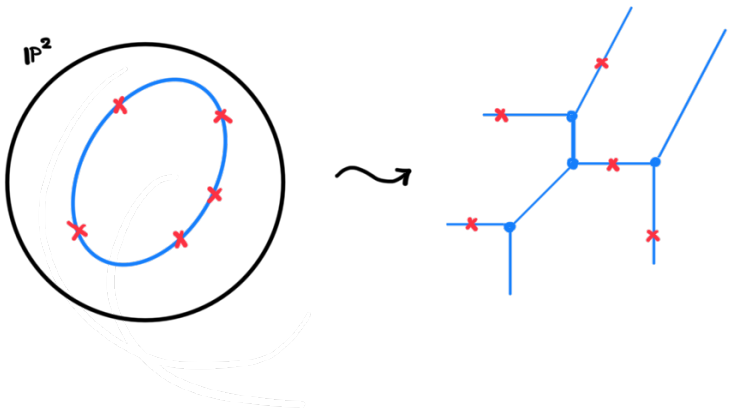
Definition 4.1. A *rational tropical curve of degree* Δ_d is given by

(i) a graph Γ of genus zero with vertices, bounded edges, and $3d$ unbounded edges,

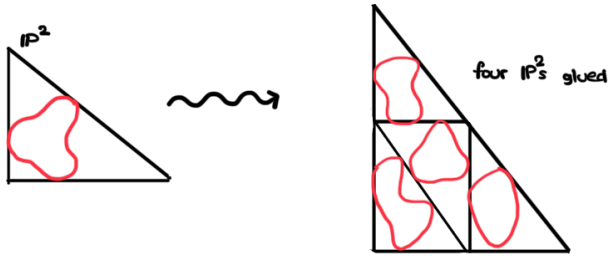
(ii) a real number for each bounded edge called the length, and

(iii) a map $h:\Gamma\to\mathbb{R}^2$ such that each unbounded edge is parallel to the corresponding vector in Δ_d and the bounded edges are mapped to segments of the specified length, and the sum of outgoing edges at each vertex is zero.

Then Milkhalkin shows that $N^{\mathbb{P}^2}_{0,d}$ is equal to the number of (weighted) rational tropical curves of degree Δ_d through $3d-1$ points. Thus for example, if $d=2$ then five points determine a unique conic as shown.



§4.1 Why should you be able to relate curve counts to tropical curves?. Of course we know that $N^{\mathbb{P}^2}_{0,d}$ is secretly a logarithmic Gromov–Witten invariant and in the logarithmic moduli space, this tropicalisation procedure produces tropical curves. The principle at play here is that invariants of a space should be related to those of its degenerations, which are precisely encoded by tropical curves.



Then the curve count we require can be computed from the counts in each segment, and these correspond to multiplicity of the vertices of the tropical curve.

§4.2 Higher genus tropical curves. Thus to compute log Gromov–Witten invariants for positive genus, we begin by asking what a higher genus generalisation of tropical curve would be. The answer comes from examining the higher genus case of K3 surfaces, which was solved earlier.

For K3 surfaces, Göttsche and Shende produce series $N_{0,h}(q)$ that evaluates the invariant $N_{0,h}$ at $q=1$. Combining this with Maulik–Pandharipande–Thomas' solution to the KKV conjecture, we have the identity

$$\sum_{g\geqslant 0} N_{g,h}u^{2g-2} = (-1)(q^{\frac{1}{2}}-q^{-\frac{1}{2}})^{-2}N_{0,h}(q) \qquad \text{at } q=e^{iu},$$

i.e. higher genus counts $N_{g,h}$ are computed by a series $N_{0,h}(q)$.

But Göttsche and Shende also give a very similar series $N^{\mathbb{P}^2}_{0,d}(q)$, so mimicking the above formula we guess that there are higher genus tropical curve counts $N^{\mathbb{P}^2}_{g,d}$ satisfying

$$\sum_{g\geqslant 0} N^{\mathbb{P}^2}_{g,d}u^{2g-2+3d} = (i(q^{\frac{1}{2}}-q^{-\frac{1}{2}}))^{3d-2}N_{0,d}(q) \qquad \text{at } q=e^{iu}.$$

Bousseau was able to give a precise definition of higher genus invariant $N^{\mathbb{P}^2}_{g,d}$ with an appropriate "λ_g condition" such that the above formula holds. Moreover in analogy with the K3 case, analogous to the K3 case, these invariants are related to higher genus tropical curve counts with refined multiplicity. The details of this are beyond the scope of these lectures.